

1. Administrative & Study Identification

Full Study Title: A Double Blind, Randomized, Placebo-Controlled Study to Evaluate the Efficacy and Safety of MK-7240/PRA023 in Subjects with Systemic Sclerosis Associated with Interstitial Lung Disease (SSc-ILD) **Short Title/Acronym:** ATHENA-SSc-ILD **Protocol Number:** PR200-104 (MK-7240-007) **NCT Number:** NCT05270668 **Sponsor Name:** Prometheus Biosciences, Inc., a subsidiary of Merck & Co., Inc. **Investigational Product:** Tulisokibart (MK-7240 / PRA023) **Phase of Development:** Phase II **Medical Monitor:** Prometheus/Merck Clinical Operations

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To obtain information on the safety and efficacy of tulisokibart (PRA023/MK-7240) compared with a placebo in adult participants with Systemic Sclerosis Associated with Interstitial Lung Disease (SSc-ILD) over a 50-week treatment period. **Secondary Objectives:** To evaluate changes in Forced Vital Capacity (FVC), high-resolution computed tomography (HRCT) QILD-WL, Revised CRISS score, Health Assessment Questionnaire Disability Index (HAQ-DI), and Living with Pulmonary Fibrosis (L-PF) QoL outcome.

2.2 Background & Rationale Systemic Sclerosis (SSc, or scleroderma) is an autoimmune disease where the immune system stimulates scarring of the skin and internal organs. SSc-associated Interstitial Lung Disease (ILD) affects over half of patients and is a leading cause of disability and death. Current therapies slow the loss of lung function but often do not improve shortness of breath, quality of life, or other physical aspects of SSc. Tulisokibart is being investigated to determine if it safely reduces inflammation and scarring, thereby slowing the progression of SSc-ILD. The optimization target is slowing the loss of lung function (FVC) and improving physical function and quality of life.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind **Randomization:** 1:1 Randomized

3.2 Planned Enrollment & Duration Target **SN\$:** Approximately 100 adult participants **Number of Sites:** Global (including the US, UK, Switzerland, and 17 other countries) **Estimated Study Start:** 28-Mar-2022 **Estimated Primary Completion:** 30-Jul-2029 **Total Study**

Duration: Up to 116 weeks (50 weeks double-blind + optional Open Label Extension + 3-visit follow-up)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years with a confirmed diagnosis of systemic sclerosis with an onset of disease ≤ 5 years ago. 2. Has diffuse cutaneous scleroderma and HRCT-confirmed SSc-related ILD. 3. FVC $\geq 45\%$ of predicted normal and Diffusing capacity of the lung for carbon monoxide (DLCO) $\geq 45\%$ of predicted normal.

4.2 Exclusion Criteria 1. Clinical evidence of an airway obstruction. 2. Women of childbearing potential not willing to use an acceptable contraceptive method. 3. Instability in background therapy dosing (e.g., nintedanib, mycophenolate mofetil [MMF], methotrexate [MTX], azathioprine, or corticosteroids).

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Tulisokibart (MK-7240/PRA023) or placebo administered throughout a 50-week treatment period via intravenous (IV) infusions every defined cycle. **Route of Administration:** Intravenous (IV) infusion. **Duration of Treatment:** 50 weeks of treatment, with an option to transition to an Open-Label Extension (OLE) period.

5.2 Concomitant & Prohibited Therapy Permitted: Stable background therapies including nintedanib, mycophenolate mofetil (MMF), methotrexate (MTX), azathioprine, and corticosteroids. **Prohibited:** Unstable doses of background therapies or other unapproved investigational agents during the trial period.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, HRCT of the lungs, Pulmonary Function Tests (PFTs), ECGs
Baseline	Day 0	Randomization, First IV Infusion
Interim	Weeks 1 to 49	Up to 17 onsite study visits for vital signs, IV infusions, QoL questionnaires, and ongoing safety monitoring. Checklists are reviewed at each scheduled interim assessment visit.
End of Study	Week 50	Final efficacy assessments (FVC, HRCT, QoL), Transition to Open-Label Extension (OLE) or 3-visit follow-up period

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Proportion of subjects reporting adverse events (AEs), serious AEs (SAEs), and AEs leading to discontinuation; along with the annual rate of change from baseline in Forced Vital Capacity (FVC) in mL over 50 weeks.

Measurement Method: Clinical safety assessments and spirometry (FVC).

7.2 Secondary & Exploratory Endpoints * **[Endpoint 1]** Change from baseline in FVC in mL at Week 50. * **[Endpoint 2]** Change from baseline in HRCT Quantitative Interstitial Lung Disease Whole Lung (QILD-WL) and Living with Pulmonary Fibrosis (L-PF) QoL Outcome at Week 50. * **[Endpoint 3]** Percentage of participants with an improvement in the Revised Composite Response Index in Systemic Sclerosis (CRISS) Score at Week 50. * **[Endpoint 4]** Change from baseline in Health Assessment Questionnaire Disability Index (HAQ-DI) at Week 50.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring of AEs, markedly abnormal laboratory values, and vitals up to Week 50.

Serious Adverse Events (SAEs): Evaluated throughout the 50-week double-blind period and optional extension; standard rapid reporting guidelines apply. **Data Monitoring Committee (DMC):**

Internal/External sponsor-designated safety committee per Prometheus Biosciences/Merck guidelines.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for primary efficacy outcomes (FVC changes). Safety Set for all AE/SAE evaluations. **Sample Size Justification:** Approximately 100 participants globally to evaluate efficacy signals and the safety profile. **Handling of Missing Data:** Standard multiple imputation or mixed-model repeated measures (MMRM) applied for longitudinal continuous data such as FVC changes.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Regular onsite and remote monitoring for protocol adherence and data integrity. **Quality Audit Mechanism:** Regular HRCT central reading and centralized spirometry review to ensure consistency in endpoint assessments. **Audit Trail:** Robust eCRF locking and query resolution handled via standard sponsor data management protocols.

11. Study Identifiers & Governance

Primary ID: PR200-104 (MK-7240-007) **Registry Number:** NCT05270668
Sponsor/Lead Organization: Prometheus Biosciences, Inc., a subsidiary of Merck & Co., Inc. **Study Title:** ATHENA-SSc-ILD
Official Regulatory Title: A Double Blind, Randomized, Placebo-Controlled Study to Evaluate the Efficacy and Safety of MK-7240/PRA023 in Subjects with Systemic Sclerosis Associated with Interstitial Lung Disease (SSc-ILD)

12. Clinical Framework (SSc-ILD Specific)

Primary Condition: Systemic Sclerosis Associated with Interstitial Lung Disease (SSc-ILD) **Diagnosis Threshold:** Onset of disease ≤ 5 years ago; FVC $\geq 45\%$, DLCO $\geq 45\%$
Medical Methodology: Investigational immunomodulatory therapy (tulisokibart) + Standard background therapies **Optimization Target:** Slowing the loss of lung function (FVC) and improving physical function/quality of life

13. Study Procedures & Methodology

Management Pathway: Intravenous (IV) infusions every defined cycle for 50 weeks **Education Protocol (HCP):** Investigator protocol adherence and IV drug administration training **Education Protocol (Patient):** Informed consent, study expectation outlining, and OLE transition counseling **Quality Audit Mechanism:** Regular HRCT central reading and centralized spirometry review

14. Observation & Follow-Up Schedule

Total Duration: Up to 116 weeks (50 weeks double-blind + optional Open Label Extension + follow-up) **Onsite Visit Cadence:** Up to 17 study visits during the 50-week treatment period **Remote Monitoring Frequency:** Periodic monitoring per standard clinical operations plan **Checklist Review Interval:** Reviewed at each scheduled interim assessment visit

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	____	[DD-MMM-YYYY]				